

BUILT ON THE CLAUDE AGENT SDK

# Marble

*"We know what's written in the Golem's mouth."*

**The world's first driverless law firm.**

Multi-agent legal design system · 10 workflows · 70 agents · 610 tests · WCAG AA  
dashboard

CONTENTS

|    |                    |    |                    |
|----|--------------------|----|--------------------|
| 01 | The Problem        | 11 | Auth & Persistence |
| 02 | The Idea           | 12 | The Session        |
| 03 | How It Works       | 13 | Permissions        |
| 04 | The Router         | 14 | Workflow Templates |
| 05 | Ten Workflows      | 15 | Design Decisions   |
| 06 | The Evaluator Gate | 16 | Using Marble       |
| 07 | The Agents         | 17 | The Numbers        |
| 08 | The Dashboard      | 18 | What's Next        |
| 09 | Document Ingestion | 19 | File Map           |
| 10 | Knowledge Base     |    |                    |

01

## Legal documents are hostile to the people who need to understand them.

Terms of service written at a graduate reading level. Contracts with risk buried in boilerplate. Privacy policies designed to obscure, not inform. And on the other side: legal questions that require expensive professional time even when the answer is straightforward.

The legal industry has a scaling problem. The work is important, the stakes are high, and there are never enough qualified people to do it.

---

02

## What if a law firm had no lawyers — only agents?

Not one agent. A team. Specialists that analyze, debate, challenge each other, transform documents, verify their own work, and learn from every engagement. A system where the agents genuinely collaborate: one proposes, another challenges, a third verifies, a human approves.

Marble is that system. Named after the act of turning rough stone into something polished and precise, it animates a team of AI legal specialists inside a workflow state machine with human gates, quality checks, and institutional memory.

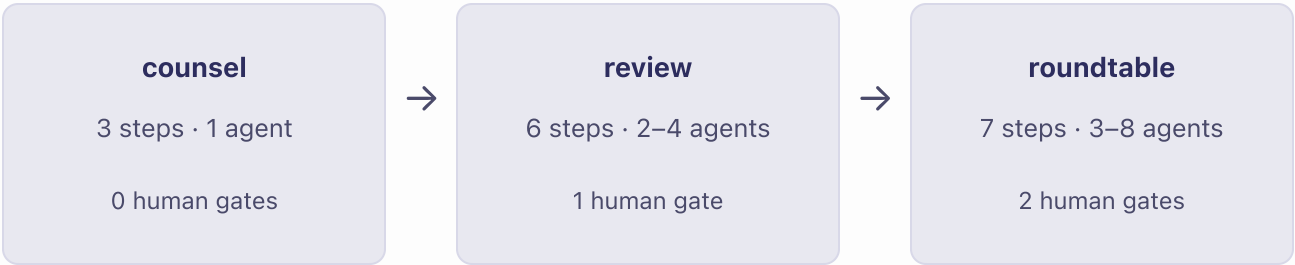
Built on the Claude Agent SDK. Built by Legit (wearelegit.ai).

03

# How It Works

A request comes in. The system reads it and asks one question:

*What is the minimum viable workflow to handle this well?*



The principle: don't use eight agents and three human gates to answer a question that needs one sentence. Default to the simplest path. Escalate complexity only when required.

04

# The Router

The highest-leverage component. A single classification determines the entire processing path.

| WHAT CAME IN                                | WHAT RUNS     |
|---------------------------------------------|---------------|
| "Redesign this document"                    | roundtable    |
| "Review this contract"                      | review        |
| "What is force majeure?"                    | counsel       |
| "Research the implications of GDPR Art. 17" | research-memo |
| "Stress-test this M&A opinion"              | adversarial   |
| Complex multi-domain engagement             | full-bench    |

Router Flags

| FLAG                                  | FIRES WHEN                             |
|---------------------------------------|----------------------------------------|
| <code>requiresEthicsFirst</code>      | Consumer data, privacy, consent        |
| <code>requiresConsistencyCheck</code> | Existing client/matter involved        |
| <code>requiresDebate</code>           | Multi-specialist, conflicting concerns |

Output Schema

```
{
  requestType: 'direct_answer' | 'single_specialist' | 'full_pipeline'
  complexity: 'low' | 'medium' | 'high'
  riskLevel: 'low' | 'medium' | 'high'
  selectedWorkflow: 'counsel' | 'review' | 'roundtable' | 'full-bench' | 'adversarial' | ...
  selectedSpecialists: string[]
  requiresDebate: boolean
  requiresEthicsFirst: boolean
  requiresConsistencyCheck: boolean
  reasoning: string
}
```

05

## Ten Workflows

Each designed for a different kind of legal work. The original three have been joined by seven more — from solo counsel sessions to full-bench adversarial reviews with red teams.

### Counsel — The Fast Path

Someone asks a legal question. One specialist answers it. No evaluator, no debate, no gates. Trust the expert and deliver. Sub-30-second response time.



*Speed is the feature. If you need checking, use Review instead.*

## Review — The Middle Path

A specialist does the analysis. An evaluator on a different model tier checks quality. A plain-language specialist translates findings into business language. A human approves before delivery. The second pair of eyes decorrelates errors.



Used for contract reviews, compliance checks, risk assessments — any task that benefits from a second pair of eyes. The evaluator produces structured output with risk scores:

| FLAG   | MEANING                                   |
|--------|-------------------------------------------|
| GREEN  | Standard terms, acceptable as-is          |
| YELLOW | Non-standard but negotiable               |
| RED    | Unfavorable or dangerous, requires action |

## Roundtable — The Full Pipeline

This is the big one. Multiple specialist agents analyze in parallel, debate their findings, challenge each other through structured exchanges, and synthesize into a unified deliverable. Multidisciplinary perspectives collide and resolve into work no single expert could produce.



## The Full Workflow Registry

Beyond the original three, seven additional workflows handle everything from pre-engagement screening to multi-team adversarial review.



| WORKFLOW        | PURPOSE                                          | TEAM SHAPE                        |
|-----------------|--------------------------------------------------|-----------------------------------|
| counsel         | Solo expert, direct answer. Sub-30-second.       | 1 specialist                      |
| review          | Specialist + evaluator with revision loop        | 2–4 agents                        |
| roundtable      | Parallel expert panel, debate, synthesis         | 3–8 specialists                   |
| full-bench      | Senior decomposition, delegated workstreams      | 8+ specialists                    |
| adversarial     | Builder + attacker + synthesizer. Stress-tested. | Specialists + red team            |
| research-memo   | Structured legal research with citations         | Researcher + evaluator + red team |
| pre-engagement  | Conflict check, KYC, engagement letter           | Intake + compliance               |
| simple-query    | Legacy: specialist + evaluator gate              | 1 specialist + evaluator          |
| contract-review | Legacy: clause-by-clause contract analysis       | 3 agents                          |
| legal-design    | Legacy: 11-step document redesign pipeline       | 8 specialists                     |

06

# The Evaluator Gate

Every workflow passes through an Evaluator Gate before delivery. This is Marble's core quality mechanism.

**The problem it solves: correlated errors.** If the same model writes an answer and checks its own answer, it makes the same mistakes both times. The writer can't see its own blind spots. You need a different model — a skeptic — to catch what the first one missed.

**Writer**  
  
Specialist (Sonnet) — Produces the answer. Executes the work. Has domain knowledge but also has blind spots.

**Checker**  
  
Evaluator (Opus) — Scores on 7 dimensions. Different model eliminates correlated confidence. Must be specific in critique.

Seven Evaluation Dimensions

| DIMENSION               | WEIGHT | WHAT IT CHECKS                          |
|-------------------------|--------|-----------------------------------------|
| Factual Correctness     | 0.20   | Are the legal claims accurate?          |
| Citation Validity       | 0.15   | Do sources exist and support claims?    |
| Policy Compliance       | 0.15   | Does output follow firm standards?      |
| Tool Consistency        | 0.10   | Do tool calls match the reasoning?      |
| Jurisdictional Accuracy | 0.15   | Right law for the right jurisdiction?   |
| Internal Consistency    | 0.15   | Does the output contradict itself?      |
| Completeness            | 0.10   | Are all parts of the request addressed? |

0.75

PASS THRESHOLD

2

MAX REVISIONS

The Evaluator doesn't grade on style — that's the plain-language specialist's job. It evaluates correctness. A wrong legal citation is a hard fail. A missing comma is not.

07

# The Agents

70 specialist agents organized across 16 practice groups — from a managing partner who assigns cases to a junior intern who does the legwork. Each agent has its own system prompt, defined role, NBA2K-style skill ratings, and model assignment. The router selects the right team for each engagement.

## FIRM LEADERSHIP

|                     |                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------|
| managing-partner    | Assigns matters, selects workflows, oversees quality. The decision-maker when practice areas conflict. |
| supervising-partner | Reviews junior work product. Ensures compliance with firm standards before delivery.                   |
| of-counsel          | Senior advisory role. Provides strategic counsel on high-stakes engagements.                           |

## DESIGN & ANALYSIS

|                           |                                                                                                                                   |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| design-reviewer           | Scores documents on 5 dimensions: readability, findability, clarity, visual design, ethics. 0–4 scale. Calculates Complexity Tax. |
| ethics-auditor            | Detects 7 categories of dark patterns. Maps to GDPR / FTC / CCPA / CPA regulations.                                               |
| service-designer          | Analyzes the full user journey. Touchpoints, tasks, emotional arcs.                                                               |
| plain-language-specialist | Cognitive load analysis. Flesch-Kincaid grade level. Plain language transformation.                                               |
| client-proxy              | Role-plays as the actual reader. Tests document against user personas.                                                            |

**TRANSFORMATION & SYNTHESIS**

---

|                                  |                                                                            |
|----------------------------------|----------------------------------------------------------------------------|
| <b>transformation-specialist</b> | Rewrites the document, preserving legal meaning while improving design.    |
| <b>meaning-guardian</b>          | Verifies no legal meaning was lost during transformation.                  |
| <b>synthesis-editor</b>          | Assembles the final dual-artifact output: redesigned doc + review package. |

**CORPORATE & TRANSACTIONAL**

---

|                             |                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------|
| <b>corporate-generalist</b> | General corporate matters. Governance, formations, board resolutions.           |
| <b>ma-specialist</b>        | Mergers and acquisitions. Due diligence, deal structuring, purchase agreements. |
| <b>contract-specialist</b>  | Contract drafting and negotiation across all commercial agreements.             |
| <b>banking-finance</b>      | Lending, structured finance, security agreements, regulatory capital.           |
| <b>capital-markets</b>      | Securities offerings, prospectus review, exchange compliance.                   |

**DISPUTES & LITIGATION**

---

|                               |                                                                                   |
|-------------------------------|-----------------------------------------------------------------------------------|
| <b>litigation-partner</b>     | Senior trial strategy. Case theory, motion practice, settlement analysis.         |
| <b>litigation-associate</b>   | Research, brief drafting, discovery management, deposition preparation.           |
| <b>arbitration-specialist</b> | International and domestic arbitration. Institutional rules, procedural strategy. |
| <b>dispute-resolution</b>     | Mediation, negotiation strategy, settlement frameworks.                           |

## REGULATORY & COMPLIANCE

---

|                             |                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------|
| <b>regulatory-counsel</b>   | Regulatory framework analysis. Licensing, reporting obligations, agency interactions. |
| <b>compliance-officer</b>   | Internal compliance programs. Policy development, training, monitoring systems.       |
| <b>antitrust-specialist</b> | Competition law. Merger clearance, cartel risk, dominance analysis.                   |
| <b>sanctions-specialist</b> | Trade sanctions, export controls, OFAC screening, embargo compliance.                 |

## SPECIALIST PRACTICE AREAS

---

|                              |                                                                            |
|------------------------------|----------------------------------------------------------------------------|
| <b>tax-counsel</b>           | Tax structuring, transfer pricing, treaty analysis, tax controversy.       |
| <b>ip-specialist</b>         | Patents, trademarks, copyright, trade secrets, licensing.                  |
| <b>privacy-counsel</b>       | GDPR, CCPA, data protection impact assessments, cross-border transfers.    |
| <b>employment-counsel</b>    | Employment contracts, workplace policies, discrimination claims, benefits. |
| <b>real-estate-counsel</b>   | Property transactions, leases, land use, zoning, title review.             |
| <b>environmental-counsel</b> | Environmental compliance, permitting, contamination liability, ESG.        |

## RISK & ADVERSARIAL

---

|                    |                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------|
| <b>risk-pricer</b> | Quantitative risk assessment. Probability-weighted exposure, financial modeling.                                 |
| <b>red-team</b>    | Adversarial challenge. Finds weaknesses in legal positions, tests arguments from opposing counsel's perspective. |

## INDUSTRY SPECIALISTS

---

|                              |                                                                                  |
|------------------------------|----------------------------------------------------------------------------------|
| <b>fintech-specialist</b>    | Financial technology regulation. Payment services, crypto, lending platforms.    |
| <b>healthcare-specialist</b> | Healthcare regulation. HIPAA, clinical trials, pharmaceutical compliance.        |
| <b>media-specialist</b>      | Media and entertainment law. Content licensing, defamation, platform regulation. |
| <b>energy-specialist</b>     | Energy regulation. Power purchase agreements, grid access, renewable incentives. |

## DESIGN & COMMUNICATION

---

|                              |                                                                                      |
|------------------------------|--------------------------------------------------------------------------------------|
| <b>ux-writer</b>             | Microcopy, interface text, error messages. Makes legal concepts navigable.           |
| <b>information-architect</b> | Document structure, navigation patterns, content hierarchy optimization.             |
| <b>visual-designer</b>       | Layout, typography, visual hierarchy. Transforms wall-of-text into scannable design. |

## QUALITY & INFRASTRUCTURE

---

|                          |                                                                                                                    |
|--------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>evaluator</b>         | Automated quality gate. Different model from specialists. 7-dimension rubric. Max 2 revision loops, then escalate. |
| <b>contract-reviewer</b> | Clause-by-clause contract analysis. Risk scores (1–5). Deviation flags. Redline drafting.                          |
| <b>legal-researcher</b>  | Case law research, statutory analysis, secondary source compilation.                                               |
| <b>project-manager</b>   | Engagement coordination, timeline tracking, resource allocation.                                                   |
| <b>knowledge-manager</b> | Curates institutional knowledge, maintains precedent libraries, extracts lessons learned.                          |

JUNIOR TEAM

|                  |                                                                              |
|------------------|------------------------------------------------------------------------------|
| junior-associate | First-pass research, document review, initial drafting under supervision.    |
| paralegal        | Document assembly, filing management, citation checking, procedural support. |
| legal-intern     | Basic research tasks, summarization, document organization.                  |

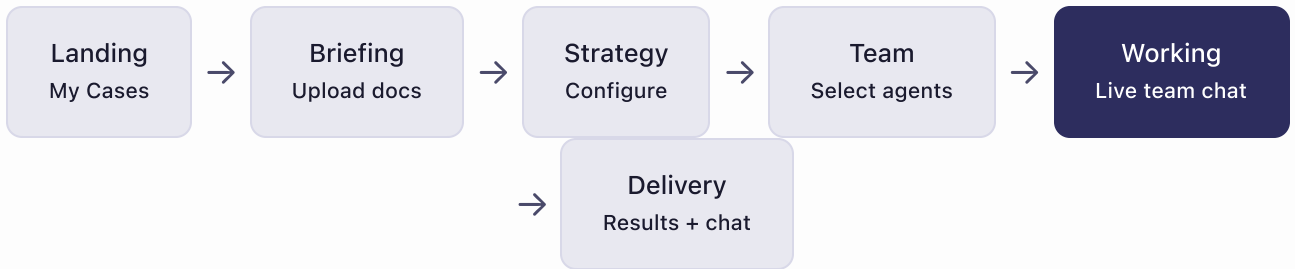
Each agent has a dedicated system prompt file (58 prompt files total, including 7 orchestrator variants for different workflow types). The router selects not just the workflow but the specific team composition — a simple NDA review might use 3 agents, while an adversarial M&A review could deploy 12.

*This isn't round-robin. It's genuine collaboration — the kind where the output is better because the agents disagree.*

08

The Dashboard

Marble has a full-stack web interface — a React single-page application with WCAG AA accessibility, responsive layouts, and an editorial design language (Inter + Cormorant Garamond, warm cream palette).



### Briefing

Upload PDFs and DOCX files, import from URLs. AI-powered briefing analysis extracts key issues and suggests workflows. Retry on failure.

### Strategy & Team

Configure engagement parameters, select workflow. Choose from team presets or hand-pick agents. Provider toggle: Claude (US) or EU Sovereign (Mistral). Offline indicator when API is unreachable.

### Working

Team chat room with real-time activity feed: agent speech bubbles, tool-use indicators, progress checklist, and reassurance messages during processing silences. Human gates appear as focus-trapped dialogs. Connection lost banner on WebSocket drop. Duplicate tab protection.

### Delivery

Six-tab editorial layout (The Work, The Story, The Scorecard, Review, Conversation, Next Steps). Derivative document generation. Loading skeleton. Responsive grids. Cowork folder save via File System Access API.

### Billing

Per-turn cost tracking with model-specific pricing (Opus, Sonnet, Haiku). Token usage breakdown. Cache read/write accounting.

### Accessibility

WCAG AA compliant: keyboard navigation with focus-visible indicators, skip-to-content, ARIA landmarks, live regions, dialog semantics, 4.5:1+ color contrast, prefers-reduced-motion support. Responsive on mobile, tablet, and desktop.

## Additional Views

My Cases (active + archived engagements), My Page (profile, custom instructions, firm personality editor), Agent Builder (NBA2K-style custom agent creator with edit mode), Marble Challenge (blind document comparison), Bet The Company (high-stakes engagement), Claw Mode (remote monitoring dashboard).



The dashboard connects to the Fastify API via authenticated fetch calls (cookie-based auth for browsers, Bearer tokens for API clients). View transitions animate navigation between screens.

09

# Document Ingestion

Legal work starts with documents. Marble has a full document ingestion pipeline that extracts text, detects structure, and makes documents available to agents.

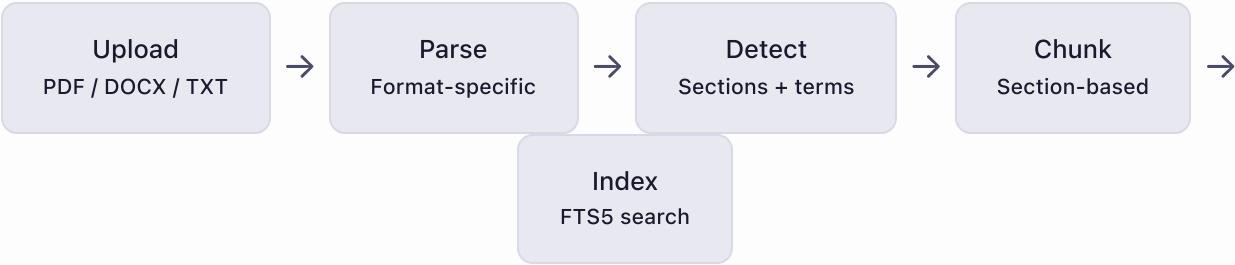
## Format Support

The parser accepts PDF (via pdf-parse), DOCX (via mammoth), and plain text formats. Each goes through format-specific extraction, then a shared structure detection pass.

## Structure Detection

After text extraction, a structure detector identifies section headings, defined terms, and tables. This transforms a flat text blob into a navigable document tree that agents can reference by section.

## The Pipeline



Documents uploaded via the dashboard briefing step flow through this pipeline automatically. The parsed output feeds into both the session (for immediate agent use) and the knowledge base (for cross-session retrieval).

---

10

## Knowledge Base

Agents need reference materials — precedents, internal policies, regulatory guidance. The knowledge base provides searchable document collections that persist across sessions.

### Indexing

Documents are chunked at section boundaries (target ~1500 words per chunk, max 2000). If no sections are detected, the fallback splits on paragraph boundaries. Each chunk is stored in SQLite with FTS5 full-text indexing.

### Retrieval

Agents query the knowledge base via MCP tools during execution. The retriever performs ranked full-text search and returns the most relevant chunks with their source document and section heading. This gives agents citable references grounded in actual uploaded materials.

### Built-in Knowledge

The system ships with curated knowledge files covering ethics audit frameworks, legal sanity checks, meaning preservation rules, pattern libraries, persona definitions, plain language guidelines, and scoring rubrics — seven knowledge domains that inform agent behavior.

11

# Auth & Persistence

The system runs on SQLite (better-sqlite3) with WAL mode for concurrent reads. Live sessions stay in memory; SQLite stores the archive.

## Dual Authentication

Two auth paths serve different clients. Browser users authenticate via email/password and receive an HttpOnly cookie ( `marble_token` ). API clients and agents use Bearer tokens ( `Authorization: Bearer shem_agent_...` ). Both paths resolve to a user identity that scopes all data access.

## Data Model

### Users

Email + bcrypt password hash. Auth tokens with expiry. User-scoped data isolation.

### Matters

Client engagements with metadata. Status tracking (active, completed, archived).

### Sessions

Workflow execution records. State snapshots, cost totals, timestamps. Linked to matters.

### Knowledge Base

FTS5-indexed document chunks. Collection-scoped. Source tracking for citations.

All API routes (11 route files) are protected by the auth middleware. Public routes (login, register) are explicitly exempted. The middleware checks cookies first, then Bearer tokens, and attaches the resolved user identity to every request.

12

## The Session

Every workflow — four steps or eleven — runs inside a session. The session is the system's memory for a single engagement.

### Events

Real-time event bus with WebSocket streaming. Routing decisions, gate results, workflow transitions, cost updates, errors.

### Audit

JSONL audit trail with SHA-256 checksum chain. Tamper-evident. Every tool call, gate decision, and cost event.

### Cost

Per-session budget tracking with hooks. Blocks tool calls when budget is exceeded.

### Memory

Institutional memory (lessons learned). Matter memory (per-client). Precedent library. Anti-pattern registry.

### Gates

Human-in-the-loop decision points. CLI (readline) or API (async resolver). Approve, reject, or modify.

### Debate

Shared debate board. Findings, challenges, responses, resolutions. RED/YELLOW/GREEN severity with confidence scores.

## Workflow

State machine with preconditions. Phase-based tool permissions enforce what agents can do at each step — even if their prompt is ignored or manipulated.

The audit trail is particularly important. Each entry's hash includes the previous entry's hash. If a single line is modified, the chain breaks — tamper-evident by design.

---

13

## Permissions

Agents can only use tools appropriate for the current workflow phase. This is enforced at the SDK level — not in the prompt, where it could be ignored.

```
canUseTool(toolName, input, options)
```

```
  Not a Marble tool? → ALLOW
```

```
  Orchestrator-only tool from subagent? → DENY
```

```
  Tool allowed in this phase? → ALLOW / DENY
```

Each workflow template defines its own permission rules. During analysis, agents can post findings but cannot challenge each other. During synthesis, no one can run verification. The system enforces temporal boundaries on what's possible.

---

14

# Workflow Templates

Every workflow is defined declaratively as a template — a data structure that says what the workflow is, not how to execute it. Templates live in a registry. The Router reads the registry. The executor reads the template.

```
interface WorkflowTemplate {  
  id: string  
  name: string  
  description: string  
  steps: string[]  
  stepDefinitions: Record<string, GenericStepDef>  
  availableTools: string[]  
  requiredAgents: string[]  
  orchestratorPrompt: string  
  phasePermissions?: Record<string, DenyRule>  
}
```

Adding a new workflow takes one file. Register it, and the Router sees it.

```
// src/workflows/templates/compliance-check.ts  
workflowRegistry.register({  
  id: 'compliance-check',  
  name: 'Compliance Check',  
  description: 'Regulatory compliance review.',  
  steps: ['intake', 'framework_selection', 'analysis',  
    'evaluator_gate', 'report', 'delivered'],  
  requiredAgents: ['compliance-specialist', 'evaluator'],  
  orchestratorPrompt: '...',  
});
```

# Design Decisions

## Why different models for specialist and evaluator?

Correlated errors. If the same model writes an answer and checks it, it makes the same mistakes both times. The #1 failure mode in multi-agent systems is correlated confidence — two instances of the same model agreeing on something that's wrong.

## Why phase-based tool permissions?

Prompt compliance is not guaranteed. An agent might ignore instructions about what tools to use. The permission system enforces tool access at the SDK level — the agent literally cannot call a blocked tool.

## Why a state machine instead of free-form agent chatter?

Legal work requires auditability. You need to prove what happened, in what order, and who approved it. A state machine with preconditions gives you that. Step 5 cannot happen before step 4. Gate decisions are recorded. The audit trail is tamper-evident.

## Why separate templates from executors?

Separation of what from how. A template says "this workflow has six steps." The executor says "here's how to run them." New workflows don't need new executors.

## Why a revision loop instead of just failing?

Most quality failures are fixable. A specialist that forgets to cite a source can add the citation. Two revision loops catches most correctable errors. After that, send it to a human.

## Why institutional memory?

The system gets better over time. When it learns that a particular type of clause is commonly missed in NDA reviews, that lesson goes into memory. The next NDA review starts with that knowledge. Every engagement teaches the system something.

## Why SQLite instead of Postgres?

Zero infrastructure. The system deploys as a single process with an embedded database file. WAL mode gives concurrent reads. FTS5 gives full-text search for the knowledge base. No connection pools, no separate database server, no Docker compose. One binary, one data directory.

## Why per-turn cost tracking?

The SDK only reports total cost at the end. But engagements can run for minutes. Users watching live agents want to see cost accumulate in real time, not discover a \$12 bill after the fact. Per-turn estimation extracts token counts from each assistant message and multiplies by model-specific rates (Opus at \$15/\$75, Sonnet at \$3/\$15, Haiku at \$0.80/\$4 per million tokens).

## Why 70 agents?

A real law firm has specialists. You would not send a tax lawyer to handle a patent dispute. The router selects the right team for each engagement — a simple NDA review might use 3 agents, while an adversarial M&A review could deploy 12. Each agent has a purpose-built system prompt tuned for its practice area. Users can also build custom agents via the Agent Builder. The cost is in prompt engineering, not compute — only the selected team runs.

---

16

# Using Marble

## CLI



```
# Counsel – quick legal question
npx tsx src/index.ts --request "What is force majeure?" \
  --workflow counsel

# Review – analyze a contract with evaluator
npx tsx src/index.ts ./nda.pdf \
  --request "Review this NDA for red flags" --budget 3.00

# Roundtable – multi-expert analysis with debate
npx tsx src/index.ts ./terms-of-service.txt \
  --workflow roundtable --audience consumer --jurisdiction EU

# Adversarial – stress-test a legal position
npx tsx src/index.ts ./opinion-letter.pdf \
  --request "Stress-test this opinion" --workflow adversarial
```

## API

```
POST /api/sessions
{
  "request": {
    "type": "review",
    "documentPath": "/path/to/contract.pdf",
    "context": { "jurisdiction": "EU" }
  },
  "options": { "budget": 5.00 }
}
```

The API returns a session ID. Connect to the WebSocket endpoint for real-time event streaming. Submit gate decisions via POST.

## Dashboard

The simplest way to use Marble: open the web dashboard, create a matter, upload your documents, select a team, and watch the agents work in real time. Gate approvals appear as interactive dialogs. Results are delivered in an editorial layout with derivative document generation.

```
# Start the server (serves both API and dashboard)
npx tsx src/api/server.ts

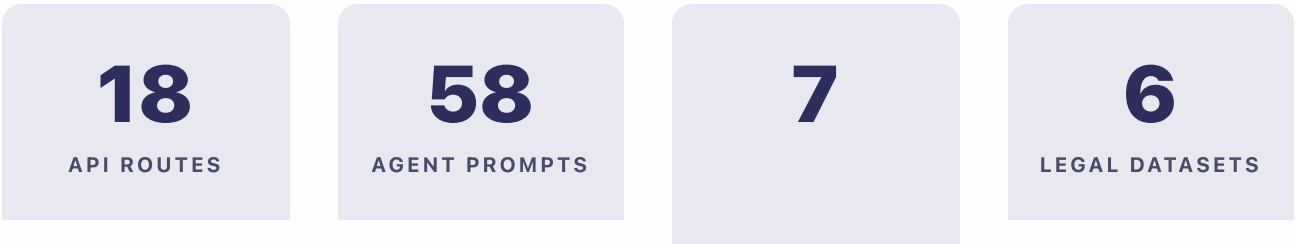
# Dashboard available at http://localhost:3000
# API at http://localhost:3000/api/*
# WebSocket at ws://localhost:3000/ws
```

17

# The Numbers



The codebase spans two packages: the backend (190 TypeScript source files) and the frontend dashboard (239 TypeScript/React source files). Both compile with zero TypeScript errors. The build produces a clean production bundle in under 2 seconds.



18

# What's Next

Most roadmap items from v1 are now shipped. Here is what was delivered and what remains.

| ITEM                                                             | STATUS  |
|------------------------------------------------------------------|---------|
| New workflow templates (7 additional)                            | SHIPPED |
| Client portal / dashboard                                        | SHIPPED |
| Document ingestion pipeline (PDF, DOCX)                          | SHIPPED |
| Auth & multi-tenancy (SQLite + user-scoped data)                 | SHIPPED |
| Knowledge base (FTS5-indexed document search + 6 legal datasets) | SHIPPED |
| Derivative document generation                                   | SHIPPED |
| Post-delivery conversation mode                                  | SHIPPED |
| Per-turn cost tracking with model-specific pricing               | SHIPPED |
| 70-agent roster across 16 practice areas                         | SHIPPED |
| WCAG AA accessibility + responsive layouts                       | SHIPPED |
| EU sovereign provider (Mistral) + ethical mode                   | SHIPPED |
| Custom agent builder with edit mode                              | SHIPPED |
| Production hardening (36 security + stability fixes)             | SHIPPED |
| Claw Mode (autonomous document processing)                       | SHIPPED |

Remaining Roadmap

| ITEM                   | DESCRIPTION                                                                               |
|------------------------|-------------------------------------------------------------------------------------------|
| Live testing & tuning  | Run real engagements end-to-end. Tune agent prompts based on actual output quality.       |
| Cross-session learning | Report card, feedback loop, and baselines are built. Wire them into the engagement cycle. |
| Multi-jurisdiction     | Parallel jurisdiction analysis within a single workflow run.                              |
| Document export        | Generate formatted PDF/DOCX outputs from delivery results.                                |
| Collaboration          | Multi-user access to matters. Comments, annotations, version history.                     |

19

# File Map

```
src/ |— dispatch.ts Universal entry point |— orchestrator.ts Legal design
executor |— config.ts Centralized env-var config | |— router/ | |— router.ts
Classification + routing | |— router-prompt.ts Router system prompt | |—
router-schema.ts Zod schema | |— workflows/ | |— registry.ts Template registry
| |— executor.ts Generic workflow executor | |— index.ts Auto-registration |
| |— templates/ 8 workflow templates | |— counsel.ts Solo expert, direct answer
(3 steps) | |— review.ts Specialist + evaluator (6 steps) | |— roundtable.ts
Parallel panel + debate (7 steps) | |— full-bench.ts Senior decomposition (7
steps) | |— adversarial.ts Builder + attacker (5 steps) | |— research-memo.ts
Research + citations (5 steps) | |— pre-engagement.ts Client onboarding (7
steps) | |— simple-query.ts Legacy: specialist + evaluator | |— contract-
review.ts Legacy: clause-by-clause | |— legal-design.ts Legacy: 11-step pipeline
| |— agents/ | |— definitions.ts All 70 agent configs | |— profiles.ts 70-
agent profile registry (skill ratings, DiceBear avatars) | |— prompts/ 58 prompt
files | |— orchestrator.ts Lead orchestrator | |— orchestrator-*.ts 4 workflow-
specific variants | |— managing-partner.ts Firm leadership | |— design-
reviewer.ts 5-dimension scorer | |— ethics-auditor.ts Dark pattern detector |
| |— litigation-partner.ts Trial strategy | |— privacy-counsel.ts GDPR / CCPA
specialist | |— red-team.ts Adversarial challenge | |— ... 50 more specialist
```

```

prompts | └─ documents/ | └─ parser.ts Document extraction entry point | └─
pdf-parser.ts PDF text extraction | └─ docx-parser.ts DOCX text extraction | └─
structure-detector.ts Section/term/table detection | └─ types.ts ParsedDocument
types | └─ knowledge-base/ | └─ indexer.ts Section-based chunking + FTS5 | └─
retriever.ts Ranked full-text search | └─ knowledge/ 7 built-in knowledge
domains | └─ ethics-audit.ts Ethics framework | └─ legal-sanity-check.ts Sanity
check rules | └─ meaning-preservation.ts Preservation rules | └─ pattern-
library.ts Design patterns | └─ persona.ts User personas | └─ plain-language.ts
Plain language guidelines | └─ scoring-rubric.ts Scoring dimensions | └─ db/ |
└─ database.ts SQLite (better-sqlite3) + migrations | └─ mcp/ | └─ server.ts
MCP server factory | └─ tools/ 18 MCP tool modules | └─ debate-board.ts Shared
agent state | └─ scoring-engine.ts Computational scoring | └─ approval-gate.ts
Human gates | └─ workflow-engine.ts 11-step state machine | └─ generic-
workflow-engine.ts N-step state machine | └─ verification-engine.ts
Self/cross/score checks | └─ evaluator-gate.ts Quality gate tools | └─ memory-
system.ts Institutional memory | └─ knowledge-base.ts Document retrieval for
agents | └─ document-reader.ts Document access | └─ pre-engagement.ts Conflict
check tools | └─ quality-check.ts Quality validation | └─ risk-pricing.ts Risk
quantification | └─ report-card.ts Quality metrics | └─ feedback-loop.ts Post-
session learning | └─ baselines.ts Quality expectations | └─ legal-md-
compiler.ts Knowledge export | └─ session-replay-testing.ts Regression tests |
└─ session/ | └─ session-state.ts Per-session state | └─ session-manager.ts
Multi-session management | └─ permissions/ | └─ dynamic-permissions.ts Phase-
based tool access | └─ hooks/ | └─ audit-logger.ts Audit trail hooks | └─
cost-tracker.ts Budget enforcement | └─ human-gate.ts Gate enforcement | └─
events/ | └─ event-bus.ts Real-time event system | └─ gates/ | └─ gate-
resolver.ts CLI + API + auto-approve | └─ types/ 11 type definition files | └─
index.ts Core types | └─ workflow.ts Workflow + template types | └─ debate.ts
Debate board types | └─ client.ts Client identity types | └─ agent-profile.ts
Agent config types | └─ engagement.ts Engagement + matter types | └─ output-
schemas.ts Agent output formats | └─ api/ | └─ server.ts Fastify API + static
serving | └─ ws-handler.ts WebSocket event streaming | └─ middleware/ | | └─
auth.ts Dual auth (cookie + Bearer) | | └─ validation.ts Zod request validation
| └─ routes/ 18 route modules | | └─ auth-routes.ts Login, register, logout | |
└─ sessions.ts Session CRUD + soul injection | | └─ engage.ts Agent engagement
(sync + webhook) | | └─ verify.ts Standalone verification | | └─ matters.ts
Matter management | | └─ briefing.ts AI briefing analysis | | └─ claw.ts Claw
Mode monitoring | | └─ challenge.ts Blind document comparison | | └─ agents.ts
Agent profiles + presets | | └─ documents.ts Document upload + parse | | └─
knowledge-base.ts KB collection management | | └─ workflows.ts Workflow template
listing | | └─ replay.ts Session replay + testing | | └─ ... 5 more (pricing,
reputation, capabilities, well-known, challenge-prompt) | └─ briefing/ | | └─
briefing-analyzer.ts AI-powered document analysis | | └─ briefing-prompt.ts
Briefing LLM prompt | | └─ briefing-schema.ts Briefing output schema | └─
derivatives/ | └─ derivative-types.ts Memo, checklist, brief types | └─
providers/ | └─ mistral.ts Mistral AI client (EU sovereign) | └─ mistral-
executor.ts Workflow execution via Mistral | └─ mistral-assembler.ts Document
assembly from Mistral | └─ tool-converter.ts MCP → Mistral tool conversion | └─

```

types.ts Provider type definitions | |— **claw/** Autonomous document processing (13 modules) | |— **watcher.ts** Filesystem watcher with debounce | |— **processor.ts** Document processing pipeline | |— **planner.ts** Budget-aware work planning | |— **registry.ts** Document tracking (SHA-256) | |— **delivery.ts** Output bundle generation | |— **local-analysis.ts** On-device analysis (Ollama) | |— **daemon.ts** macOS LaunchAgent management | |— ... 6 more modules | |— **assembly/** | |— ... Document assembly (HTML, DOCX) | |— **utils/** | |— **stream-messages.ts** Message streaming + per-turn cost | |— **audit-persistence.ts** JSONL + checksum chain | |— **error-recovery.ts** Structured error handling | |— **logger.ts** Structured logging utility | |— **fs-helpers.ts** Atomic writes + backup recovery | |— **index.ts** CLI entry point **viz/src/** React dashboard (239 source files) |— **App.tsx** Root component + view routing + transitions |— **auth/** Login/register UI |— **landing/** Landing page, QuickStart, YOLO |— **my-cases/** Case management + archives |— **my-page/** Profile, custom instructions, soul editor |— **briefing/** Document upload + analysis + retry |— **staffing/** Strategy config, team selection, provider toggle |— **working/** Team chat room + real-time checklist |— **components/** ActivityCard, ReassuranceCard, ProgressSidebar |— **hooks/** useWorkingState, useReassuranceInjector | |— **utils/** toolLabels |— **delivery/** Results + derivatives + skeleton loading |— **agent-builder/** Custom agent creator (3-step wizard + edit) |— **challenge/** Blind document comparison |— **bet-the-company/** High-stakes engagement view |— **claw/** Remote monitoring dashboard |— **cowork/** File System Access API saves |— **hooks/** useMediaQuery, useTabLock |— **components/** GateDialog (focus trap), ErrorToast, MarbleMark |— **archive/** Knowledge base search UI

# Marble

610 tests · 70 agents · 10 workflows · WCAG AA · 0 TS errors

*"We know what's written in the Golem's mouth."*